WORKED SOLUTION

STEP I 2013, Question 8

Pure • Composite Functions

> **STEP I 2013, Question 8**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/13-s1-q8

The key idea

A composite pq means "do q first, then p", and it is only defined where the output of q lands inside the domain of p. So the domain comes from q (trimmed if necessary), and the range is found by asking which values p actually receives. The algebra is light; the marks are in being careful with $\sqrt{x^2} = |x|$ and with the domains. In part (iii) the identity $(x + \sqrt{x^2-1})(x - \sqrt{x^2-1}) = 1$ does most of the work.

Solution

Part (i)

- $\mathrm{cb}(x) = \mathrm{c}(\ln x) = 2\ln x$. The domain is that of b, namely $x > 0$, and c accepts every real number. As $\ln x$ takes every real value, so does $2\ln x$: the range is $\mathbb{R}$.
- $\mathrm{ab}(x) = (\ln x)^2$, with domain $x > 0$. Since $\ln x$ takes every real value, its square takes every value in $[0, \infty)$ (with 0 at $x = 1$): the range is $y \geq 0$.
- $\mathrm{da}(x) = \sqrt{x^2} = |x|$, not x, because the square root sign means the non-negative root. Here $x^2 \geq 0$ always, so the domain is all of $\mathbb{R}$, and the range is $y \geq 0$.
- $\mathrm{ad}(x) = (\sqrt{x})^2 = x$, but only for $x \geq 0$, since d must act first. So the domain is $x \geq 0$ and the range is $y \geq 0$.

Notice that da and ad have the same range but different domains and different formulae: they are different functions.

Part (ii)

For fg: $\mathrm{g}(x) = \sqrt{x^2+1} \geq 1$ for every real x, so $\mathrm{g}(x)$ always lies in the domain of f. Then

$$\mathrm{fg}(x) = \sqrt{(x^2+1)-1} = \sqrt{x^2} = |x|, \qquad \text{domain } \mathbb{R}, \quad \text{range } y \geq 0.$$

For gf: f acts first, so we need $|x| \geq 1$; and g accepts any input. Then

$$\mathrm{gf}(x) = \sqrt{(x^2-1)+1} = |x|, \qquad \text{domain } |x| \geq 1, \quad \text{range } y \geq 1.$$

Again the two formulae agree, but the functions differ: the domain of gf excludes $-1 < x < 1$, and so its range starts at 1.

Part (iii)

The basic fact is

$$\left(x + \sqrt{x^2-1}\right)\left(x - \sqrt{x^2-1}\right) = x^2 - (x^2-1) = 1,$$

so $\mathrm{h}(x)\,\mathrm{k}(x) = 1$ for $x \geq 1$.

The graph of h ($x \geq 1$).

- $\mathrm{h}(1) = 1$.
- For $x > 1$, $\mathrm{h}'(x) = 1 + \dfrac{x}{\sqrt{x^2-1}} > 0$, so h is increasing; and $\mathrm{h}'(x) \to \infty$ as $x \to 1^+$, so the curve leaves $(1,1)$ with a vertical tangent.
- As $x \to \infty$, $\mathrm{h}(x) \to \infty$. For the asymptote,

  $$\mathrm{h}(x) - 2x = \sqrt{x^2-1} - x = \frac{-1}{\sqrt{x^2-1}+x} \to 0^-,$$

  so $y = 2x$ is an asymptote, approached from below.

So h increases continuously from 1 without bound: its range is $[1, \infty)$.

The graph of k, branch $x \geq 1$.

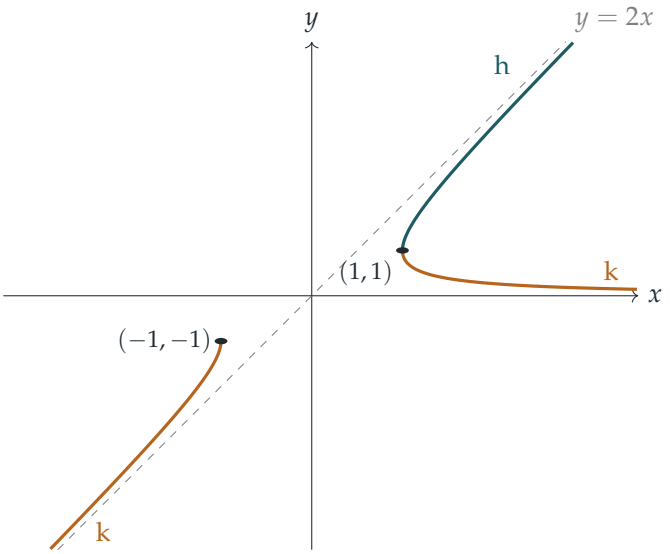
- $\mathrm{k}(1) = 1$ and, by the basic fact, $\mathrm{k}(x) = 1/\mathrm{h}(x)$.
- Since h is positive and increasing to infinity, k is positive and decreasing, with $\mathrm{k}(x) \to 0^+$. So $y = 0$ is an asymptote, approached from above.
- $\mathrm{k}'(x) = 1 - \dfrac{x}{\sqrt{x^2-1}} \to -\infty$ as $x \to 1^+$: a vertical tangent at $(1,1)$ again, where this branch joins the graph of h.

The graph of k, branch $x \leq -1$.

- $\mathrm{k}(-1) = -1$.
- For $x < -1$ we have $x < 0$, so $\mathrm{k}'(x) = 1 - \dfrac{x}{\sqrt{x^2-1}} > 1 > 0$: k is increasing. As $x \to -1^-$, $\mathrm{k}'(x) \to +\infty$, so there is a vertical tangent at $(-1,-1)$.

- Both x and $-\sqrt{x^2-1}$ are negative, so $\mathrm{k}(x) \to -\infty$ as $x \to -\infty$. Writing $x = -u$ with $u \geq 1$,
$$\mathrm{k}(x) - 2x = u - \sqrt{u^2-1} = \frac{1}{u + \sqrt{u^2-1}} \to 0^+,$$
so $y = 2x$ is an asymptote of this branch too, approached from above.

This branch takes every value in $(-\infty, -1]$.



(A neat way to see all this at once: put $x = \cosh t$ with $t \geq 0$. Then $\sqrt{x^2-1} = \sinh t$, so $\mathrm{h}(x) = e^t$ and $\mathrm{k}(x) = e^{-t}$; for $x = -\cosh t$, $\mathrm{k}(x) = -e^t$.)

The composite kh. The domain of h is $x \geq 1$, and every value $\mathrm{h}(x)$ satisfies $\mathrm{h}(x) \geq 1$, which lies in the domain of k. So kh is defined on all of $x \geq 1$.

? **Ask yourself** Which part of the graph of k does kh actually use?

Only the branch $x \geq 1$, since h never outputs anything below 1. Moreover h takes *every* value in $[1, \infty)$, so kh takes exactly the values of k on $[1, \infty)$. On that branch k decreases continuously from $\mathrm{k}(1) = 1$ towards 0, never reaching 0 (because $\mathrm{k} = 1/\mathrm{h} > 0$). Hence

$$\mathrm{kh}: \quad \text{domain } x \geq 1, \qquad \text{range } 0 < y \leq 1.$$

In the hyperbolic picture, $\mathrm{kh}(\cosh t) = \mathrm{k}(e^t)$, which is a decreasing function of t starting at 1 when $t = 0$.

Checks

- Spot values: $\mathrm{h}(2) = 2 + \sqrt{3} \approx 3.732$ and $\mathrm{k}(2) = 2 - \sqrt{3} \approx 0.268$; their product is 1. Also $\mathrm{k}(-2) = -2 - \sqrt{3} \approx -3.732$, which is just above $2x = -4$, as the asymptote

analysis predicts.

- $\mathrm{kh}(1) = \mathrm{k}(1) = 1$, and $\mathrm{kh}(2) = \mathrm{k}(3.732\ldots) = 3.732 - \sqrt{3.732^2 - 1} \approx 0.136$, inside $(0, 1]$.
- At $x = 1000$, $\mathrm{h}(x) - 2x \approx -0.0005$ and $\mathrm{k}(x) \approx 0.0005$, matching $-1/(2x)$ and $1/(2x)$.

Common slips

- Writing $\sqrt{x^2} = x$. It is $|x|$, which is why da, fg and gf are not the identity.
- Giving the domain of gf as $\mathbb{R}$ because the final formula $|x|$ makes sense everywhere. The domain of a composite is fixed by the first function applied.
- Missing the second branch of k for $x \leq -1$, or its asymptote $y = 2x$.

Beyond STEP

At university a function is a triple: a domain, a codomain and a rule. Two functions with the same rule but different domains, like fg and gf here, are genuinely different, and this precision matters as soon as you discuss inverses, injectivity and surjectivity. The function h is $e^{\operatorname{arcosh} x}$, and $\ln \mathrm{h}(x)$ is the standard formula for $\operatorname{arcosh} x$. Abbott's *Understanding Analysis* treats functions, continuity and monotonicity with this kind of care.

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